Sensitivity Analysis of Convective Parameterization Schemes (CPS) and Planetary Boundary Layer (PBL) on WRF Simulations of Summer Precipitation in East Asia for 2022

Junseo Park[1], Taeho Mun[2], Eunji Kim[3], Haeun Jo[4], and Dong-Hyun Cha[5]

[1] School of Urban and Environmental Engineering, Ulsan National Institute of Science & Technology (UNIST), Ulsan, Republic of Korea *junseo1026@unist.ac.kr*
[2] School of Urban and Environmental Engineering, Ulsan National Institute of Science & Technology (UNIST), Ulsan, Republic of Korea
[3] School of Urban and Environmental Engineering, Ulsan National Institute of Science & Technology (UNIST), Ulsan, Republic of Korea
[4] School of Urban and Environmental Engineering, Ulsan National Institute of Science & Technology (UNIST), Ulsan, Republic of Korea
[5] School of Urban and Environmental Engineering, Ulsan National Institute of Science & Technology (UNIST), Ulsan, Republic of Korea

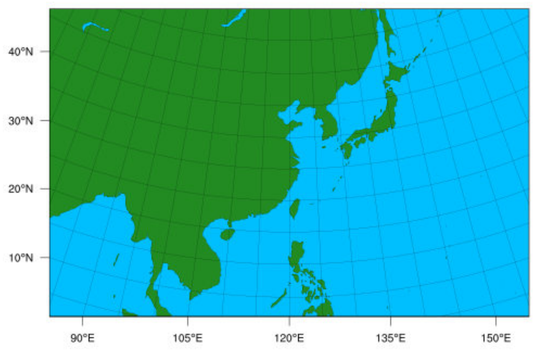
KSCC 2023-2 Poster Session : Climate Change Science 2 (P-14)

1. Introduction

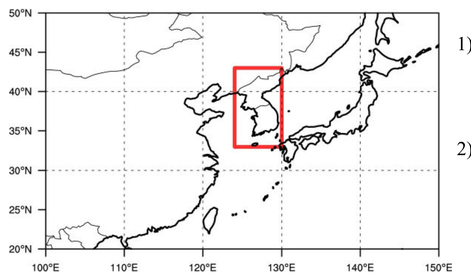
1. In East Asia, around half of the yearly rainfall occurs in summer, and predicting the **East Asian summer monsoon (EASM)** is challenging due to various influencing factors.
2. Understanding East Asian summer precipitation and enhancing simulation accuracy through optimal experimental design is important.
3. Summer precipitation in East Asia is mainly linked to convective precipitation. The complex topography of East Asia affects heat, moisture, and momentum exchange between land, ocean, and atmosphere, along with boundary mixing processes.
4. Although 2022 summer precipitation in East Asia wasn't notably different from the average, the Korean Peninsula experienced an unusual form, breaking the maximum hourly precipitation limit. Southeastern China recorded the second-highest precipitation since 1961 from June to September 2022.
5. This study used the **Weather Research and Forecasting (WRF)** model to analyze 2022 East Asia summer precipitation simulations based on different **PBL** schemes and **CPS**.
6. The PBL is crucial for understanding the interaction between convection and horizontal flow. This study used PBL methods, which are **Yonsei University (YSU)** and **Asymmetric Convective Model (ACM2).**
7. Comparing the impact of **KIM Simplified Arakawa-Schubert (KSAS)** and **Multi-scale Kain-Fritsch (MSKF)** CPS on precipitation amounts prediction.

2. Data & Methods

• Model Domain



• Analysis Domain



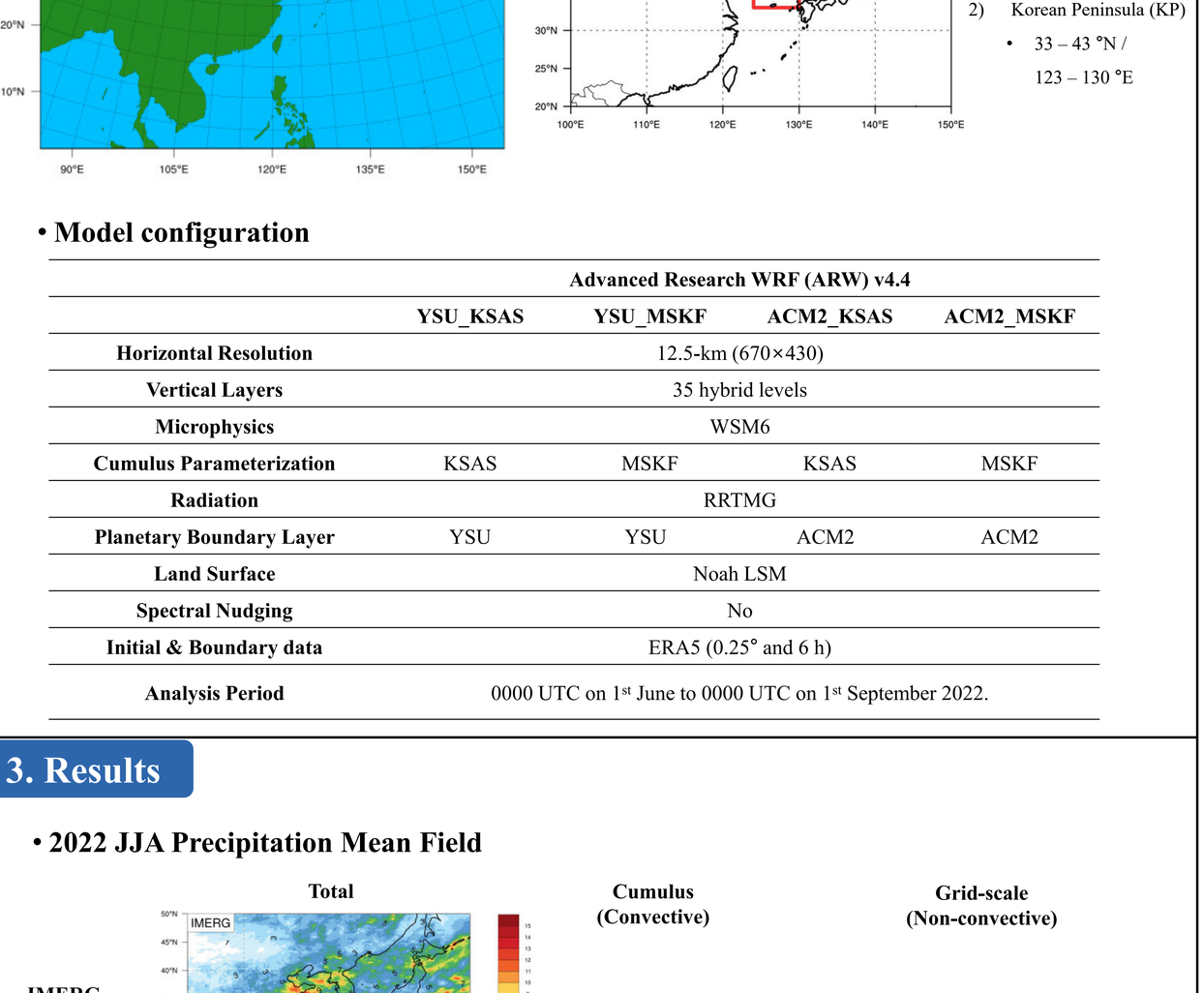
1) East Asia (EA)
   - 20 – 50 °N / 100 – 150 °E
2) Korean Peninsula (KP)
   - 33 – 43 °N / 123 – 130 °E

• Model configuration

	Advanced Research WRF (ARW) v4.4			
	YSU_KSAS	**YSU_MSKF**	**ACM2_KSAS**	**ACM2_MSKF**
Horizontal Resolution	12.5-km (670×430)			
Vertical Layers	35 hybrid levels			
Microphysics	WSM6			
Cumulus Parameterization	KSAS	MSKF	KSAS	MSKF
Radiation	RRTMG			
Planetary Boundary Layer	YSU	YSU	ACM2	ACM2
Land Surface	Noah LSM			
Spectral Nudging	No			
Initial & Boundary data	ERA5 (0.25° and 6 h)			
Analysis Period	0000 UTC on 1st June to 0000 UTC on 1st September 2022.			

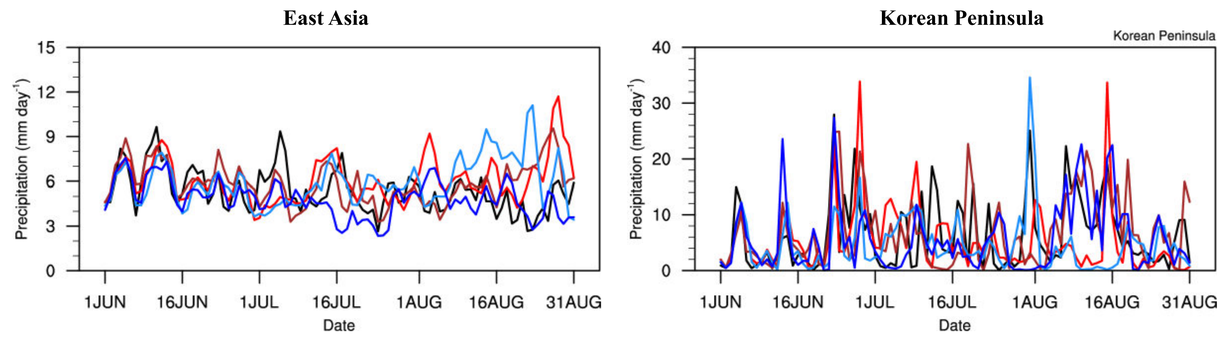
3. Results

• 2022 JJA Precipitation Mean Field



- The KSAS scheme is known for addressing most instabilities in CPS, but not on Micro-Physics Schemes (MPS). Strong convective precipitation resulted in weak grid-scale precipitation, and the simulation did not capture strong precipitation accurately. Both KSAS scheme experiments simulated a large amount of convective precipitation compared to non-convective precipitation.
- In contrast, both MSKF scheme experiments simulated stronger non-convective precipitation and weaker convective precipitation.
- The **YSU** scheme appears to overestimate coastal precipitation, and underestimate ocean precipitation. **YSU_KSAS**, however, performed well in simulating the Korean Peninsula region.
- The **ACM2_KSAS** experiment simulated heavy precipitation in the northwest Pacific, likely influenced by the PBL height and the northwest subtropical Pacific high.
- The **ACM2_MSKF** exhibited a similar spatial distribution of precipitation but almost failed to simulate precipitation in the Philippine Sea.
- Overall, the **YSU_KSAS** provided the most accurate estimation of average precipitation in the both of East Asia and Korean Peninsula regions.
- It appears that the **sensitivity of CPS in simulations of summer precipitation is more significant than that of PBL**.

• Time Series – 2022 JJA



▪ Statistics – 2022 JJA

- **East Asia**

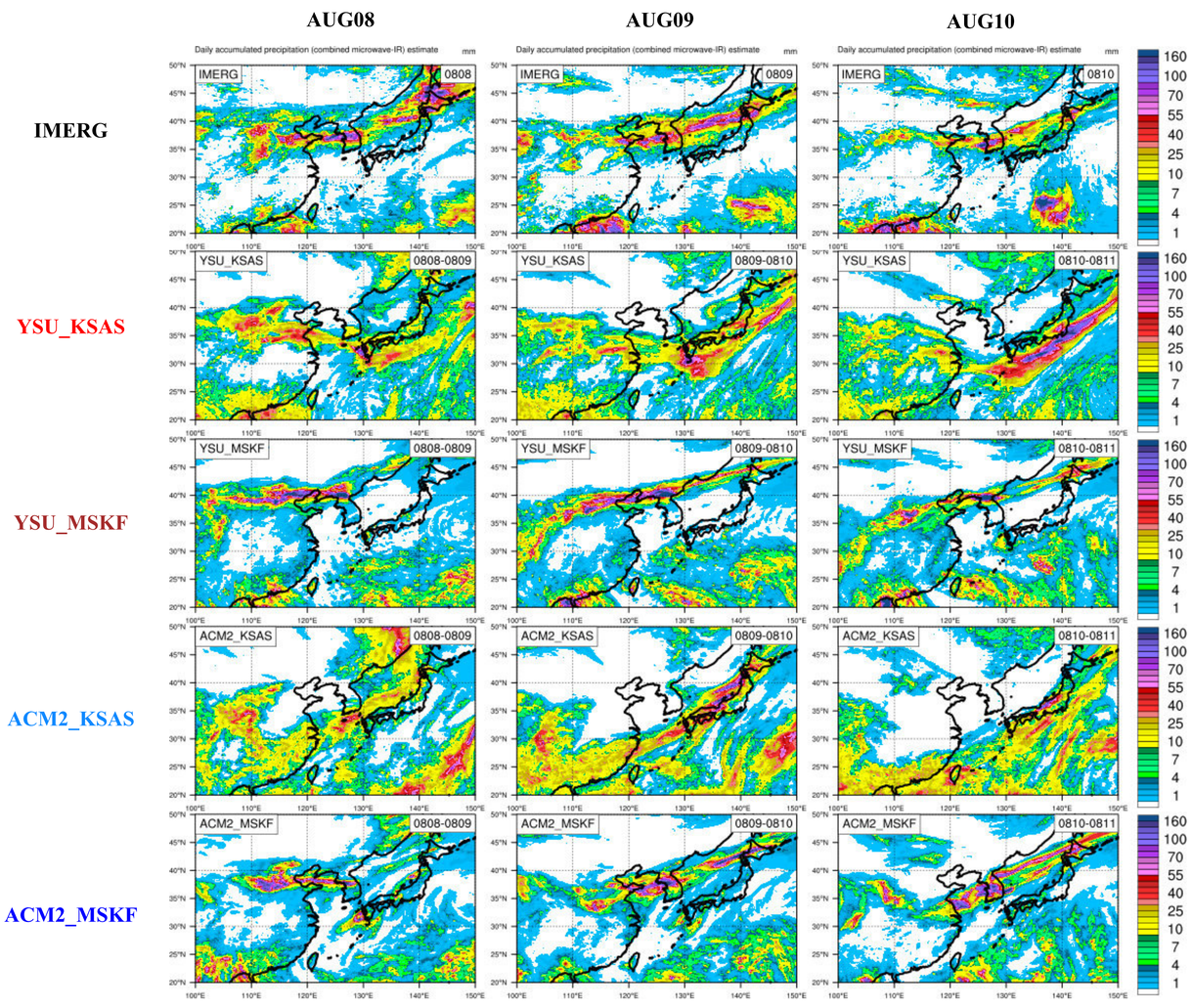
	IMERG	YSU_KSAS	YSU_MSKF	ACM2_KSAS	ACM2_MSKF
Mean (mm day⁻¹)	5.35	6.08	5.86	5.96	4.82
Bias (mm day⁻¹)		0.73	0.50	0.61	-0.53
RMSE (mm day⁻¹)		**2.84**	3.81	4.16	3.29
SCORR		**0.91**	0.84	0.82	0.84

- **Korean Peninsula**

	IMERG	YSU_KSAS	YSU_MSKF	ACM2_KSAS	ACM2_MSKF
Mean (mm day⁻¹)	6.29	5.35	6.92	4.67	5.60
Bias (mm day⁻¹)		-0.94	0.63	-1.62	-0.68
RMSE (mm day⁻¹)		**2.22**	3.80	2.73	2.80
SCORR		**0.95**	0.87	0.92	0.90

- In the East Asia domain, every scheme combination except for ACM2_MSKF overestimated daily precipitation compared to IMERG. (YSU_KSAS > ACM2_KSAS > YSU_MSKF > **IMERG** > ACM2_MSKF)
- In the later stages, experiments applying the KSAS CPS tend to simulate higher daily average precipitation compared to observations.
- On the Korean Peninsula, only YSU_MSKF overestimated the daily precipitation compared to IMERG. (YSU_MSKF > **IMERG** > ACM2_MSKF > YSU_KSAS > ACM2_KSAS)
- **YSU_KSAS** shows the highest spatial correlation with IMERG in both the East Asia and Korean Peninsula domains.
- There is no continuous relationship among the four experiments within each domain.

• Extreme Precipitation Case (AUG 8th – 10th) – Precipitation Mean Field



▪ Statistics – 2022 AUG 8th – 10th

- **East Asia**

	IMERG	YSU_KSAS	YSU_MSKF	ACM2_KSAS	ACM2_MSKF
Mean (mm day⁻¹)	17.43	19.29	16.83	22.15	14.31
Bias (mm day⁻¹)		1.86	-0.61	4.71	-3.13
RMSE (mm day⁻¹)		42.49	48.45	40.81	**34.66**
SCORR		0.28	0.30	0.34	**0.50**

- **Korean Peninsula**

	IMERG	YSU_KSAS	YSU_MSKF	ACM2_KSAS	ACM2_MSKF
Mean (mm day⁻¹)	43.31	7.44	46.61	7.16	47.23
Bias (mm day⁻¹)		-35.87	3.30	-36.15	3.92
RMSE (mm day⁻¹)		71.98	139.81	72.99	**48.91**
SCORR		0.17	0.10	0.10	**0.80**

- Contrary to the overall analysis period, the extreme precipitation case exhibits some conflicting results.
- Both **MSKF** experiments well simulated long narrow front-shaped precipitation around 40°N.
- However, both **KSAS** experiments did not simulate long continuous front shape and are showing precipitation over a wide region.
- It seems that **KSAS** shows more equatorward front precipitation, and **MSKF** showing more poleward front precipitation.
- Consequently, the **ACM2_MSKF** shows the highest spatial correlation in both the East Asia and Korean Peninsula domains.
- Still there are no continuous relationship among the four experiments within each domain.

4. Summary & Future Work

1. **KSAS CPS** addressed instabilities but struggled with strong precipitation.
2. **MSKF CPS** showed strength in non-convective precipitation but weakness in convective events.
3. PBL schemes (YSU and ACM2) impacted precipitation patterns, with YSU overestimating coastal and underestimating ocean precipitation.
4. **YSU_KSAS** consistently provided the **most accurate average precipitation estimation** in East Asia and the Korean Peninsula on overall period.
5. **ACM2_MSKF** showed the highest spatial correlation in East Asia and the Korean Peninsula on **extreme precipitation case**.

1. Investigate the performance of other CPS and PBL schemes to understand their interactions and applicability under diverse atmospheric conditions.
2. Assess the sensitivity of the model to different MPS to refine precipitation simulations.
3. Conduct in-depth research on extreme precipitation events to understand the underlying mechanisms and factors influencing such occurrences.
4. Continuously refine regional climate models to enhance predictions and preparedness in the face of **changing climatic conditions**.

✓ *이 연구는 정부(과학기술정보통신부)의 재원으로 한국연구재단의 지원을 받아 수행된 연구임(NRF-2022R1F1A1076381).*